

DEEPTHI VENMANASSERIL JAINLAL

[LinkedIn](#) | deepthivj96@gmail.com | +1 (341) 252-9234 | [GitHub](#) | [Medium](#) | [Portfolio](#) | 4.7 Years Exp

PROFESSIONAL SUMMARY

ML Engineer and Data Scientist with a deep enthusiasm for finding patterns in complex data and building systems that drive real impact. Enthusiastic about Machine Learning, Deep Learning, and Generative AI, with experience in developing and deploying end-to-end solutions that solve business problems. Competent in working individually and in a team to deliver solutions that scale. Committed to rigorous thinking, clear communication, and continuous learning.

CORE COMPETENCIES

ML & Deep Learning: PyTorch · CUDA · CNN · YOLOv3 · Transformers · LLMs · RAG · Fine-tuning

Agentic & GenAI: LangChain · LangGraph · Gemini API · Vertex AI · Agentic AI Workflows, Prompt Engineering, LLM Evaluation

MLOps & Infra: Docker · CI/CD · Cloud Run · Cloud Build · GCP · BigQuery · Dataform · Databricks · Azure

Languages & Tools: Python · C# · gRPC · REST APIs · SQL · Alteryx · Qlik Sense · PowerBI · Pyspark

Classical ML: XGBoost · Random Forest · scikit-learn · Anomaly Detection · Gurobi · Statistical Modeling

PROFESSIONAL EXPERIENCE

Data Scientist – ML, MLOps & Supply chain Analytics | **Ford Motor Company** | Aug 2023 – Jul 2025 | Hybrid

- Built and deployed Total Landed Cost Estimation model (linear regression, Linear mixed effect model BigQuery/Dataform/Alteryx) enabling \$16M in supplier cost savings with automated **ETL, data quality validation**, and scheduled orchestration in BigQuery/Dataform
- Developed Same-Like Part Dashboard with percentile-based anomaly detection to flag pricing inconsistencies across part families
- Improved Part Price Prediction pipeline (Random Forest) from 76% to 84% accuracy, driving \$5M in savings
- Developed and Deployed GenAI amortization detection pipeline (LLMs on Vertex AI, GCP) — identified **\$127.8M** in total opportunities, with **\$53.5M audit-confirmed** using prompt engineering, validation checks, and hallucination guardrails
- Productionized ML models via Cloud Run, Cloud Build, and scheduled BigQuery pipelines using MLOps best practices

Software Engineer / ML Engineer | **Philips Healthcare** | Jul 2021 – Jul 2023 | Hybrid

- Built risk prediction model (Random Forest) and synthetic data augmentation pipeline (linear regression) for risk datasets
- Applied Monte Carlo simulation on risk graph by randomizing likelihood and impact values to analyse risk exposure
- Developed patient monitor simulator desktop apps (C#, WPF, gRPC) and test automation frameworks (Gherkin, C#)

Data Science Research Intern | **Philips Healthcare** | Sep 2020 – Jul 2021 | India

- Trained nnU-Net in PyTorch on 900+ annotated medical images, achieving 89% detection accuracy along with CUDA GPU optimization.
- Defined Ground truth validation using Dice score and IOU Score.
- Annotated the images and created masks using python code for training by consulting doctor for gaining knowledge on medical image features

KEY PROJECTS

- Agentic Vehicle Inspection AI: Multi-agent LangGraph + Gemini 2.5 system with CV defect detection, LLM reasoning, and structured report generation
- MedMat — Top 6 Finalist, Stanford × Google DeepMind Hackathon: AI platform for finding nearest suppliers and safety datasheet documentation using Google AI Studio
- SurgiPlan AI — NVIDIA Cosmos Hackathon: Physical AI surgical robot simulation on Nebius H100 GPUs; full-stack Flask deployment

EDUCATION

MS, Engineering Management (GPA: 3.33) — Westcliff University, CA | Oct 2025 – Oct 2027 | Business Analytics · Risk Management. Project Management

MTech, Computer Science — VIT University, India | Aug 2019 – Jul 2021 | Algorithms · AI/ML · Statistics · Bayesian Inference

EVIDENCE OF EXCELLENCE

- 🏆 Top 6 Finalist — Stanford × Google DeepMind Hackathon — MedMat selected from hundreds of global teams
- 🏆 Recognized at Ford Motor Company — Recognized for supply chain analytics excellence and cross-functional collaboration